Quiz 2 formula sheet

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CH 7 RL switching · find current and voltage

Before and after the switch

At DC steady state short L and open C. Solve the old circuit for I_0 and V_0, then the new circuit for the final values.

$$V_0 = V_{R_x} = I_{R_x} R_x = V_s \frac{R_x}{R_s + R_x}$$

Above: C spans the same two nodes as Rx in the DC divider. In 8.4, V_0 = 10(6)/(4 + 6) = 6 V.

$$i_L(0^+) = i_L(0^-) = I_0, \quad v_C(0^+) = v_C(0^-) = V_0$$

$$R_1 \parallel R_2 = \frac{R_1 R_2}{R_1 + R_2}, \qquad \tau = \frac{L}{R_{\text{Th}}}$$

Find RTh looking into the inductor terminals after switching. Short independent voltage sources and open independent current sources. Trace the discharge path.

Source removed or still connected?

$$i_L(t) = i_L(\infty) + [I_0 - i_L(\infty)]e^{-t/\tau}$$

Source-free: $i_L(t) = I_0 e^{-t/\tau}$

$$v_L = L\, di_L/dt, \qquad v_R = R i_R$$

Series DC: $i_L(\infty) = V_s/R, \quad v_L = V_s - R i_L$

Current enters the + terminal. At a later switch restart with the current at that instant and elapsed time. At steady DC, vL = 0.

CH 8 RLC · choose the current or voltage being asked for

Circuit constants and initial slopes

$$\alpha_s = R/(2L), \quad \alpha_p = 1/(2RC), \quad \omega_0 = 1/\sqrt{LC}$$

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}, \quad \omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

Overdamped α > ω_0 · Critical α = ω_0 · Underdamped α < ω_0.

$$i_L'(0^+) = v_L(0^+)/L, \quad v_C'(0^+) = i_C(0^+)/C$$

Use KVL or KCL to get the starting slope

8.4: $i'(0^+) = \frac{V_0 - R I_0}{L} = \frac{6 - 9(1)}{0.5} = -6\ \text{A/s}$

Above uses 8.4's capacitor polarity. Read the signs from the circuit.

Source-free parallel: $v'(0^+) = -\frac{V_0/R + I_0}{C}$

Parallel branch currents downward, v positive at top. For current injected by a source, add Is/C to v′(0⁺).

Source-free series · current i(t)

Overdamped

$$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

$$A_1 + A_2 = I_0, \quad s_1 A_1 + s_2 A_2 = i'(0^+)$$

Critically damped · her coefficient order

$$i(t) = (A_2 + A_1 t)e^{-\alpha t}$$

$$A_2 = I_0, \quad A_1 = i'(0^+) + \alpha I_0$$

Underdamped

$$i(t) = e^{-\alpha t}(B_1 \cos \omega_d t + B_2 \sin \omega_d t)$$

$$B_1 = I_0, \quad B_2 = \frac{i'(0^+) + \alpha I_0}{\omega_d}$$

Source-free · capacitor voltage v(t)

Parallel uses αp. For series capacitor voltage (PS7 Q2), use αs.

$$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} \quad \text{(overdamped)}$$

$$A_1 + A_2 = V_0, \quad s_1 A_1 + s_2 A_2 = v'(0^+)$$

$$v(t) = (A_1 + A_2 t)e^{-\alpha t} \quad \text{(critical)}$$

$$A_1 = V_0, \quad A_2 = v'(0^+) + \alpha V_0$$

$$v(t) = e^{-\alpha t}(A_1 \cos \omega_d t + A_2 \sin \omega_d t)$$

$$A_1 = V_0, \quad A_2 = \frac{v'(0^+) + \alpha V_0}{\omega_d} \quad \text{(under)}$$

Series voltage step · capacitor voltage

$$v_C(t) = V_s + A_1 e^{s_1 t} + A_2 e^{s_2 t} \quad \text{(over)}$$

$$A_1 + A_2 = V_0 - V_s, \; s_1 A_1 + s_2 A_2 = v_C'(0^+)$$

$$v_C(t) = V_s + (A_1 + A_2 t)e^{-\alpha t} \quad \text{(critical)}$$

$$A_1 = V_0 - V_s, \quad A_2 = v_C'(0^+) + \alpha A_1$$

$$v_C(t) = V_s + e^{-\alpha t}(A_1 \cos \omega_d t + A_2 \sin \omega_d t)$$

$$A_1 = V_0 - V_s, \quad A_2 = \frac{v_C'(0^+) + \alpha A_1}{\omega_d} \quad \text{(under)}$$

With i entering C's + terminal, vC′(0⁺) = I_0/C and i(t) = C vC′(t).

Parallel current step · inductor current

$$i_L(t) = I_s + A_1 e^{s_1 t} + A_2 e^{s_2 t} \quad \text{(over)}$$

$$A_1 + A_2 = I_0 - I_s, \; s_1 A_1 + s_2 A_2 = i_L'(0^+)$$

$$i_L(t) = I_s + (A_1 + A_2 t)e^{-\alpha t} \quad \text{(critical)}$$

$$A_1 = I_0 - I_s, \quad A_2 = i_L'(0^+) + \alpha A_1$$

$$i_L(t) = I_s + e^{-\alpha t}(A_1 \cos \omega_d t + A_2 \sin \omega_d t)$$

$$A_1 = I_0 - I_s, \quad A_2 = \frac{i_L'(0^+) + \alpha A_1}{\omega_d} \quad \text{(under)}$$

With iL entering L's + terminal, iL′(0⁺) = V_0/L and v(t) = L iL′(t).

CH 9 AC voltage and current · cosine phasors

Convert the source and the components

$$v(t) = V_m \cos(\omega t + \phi) \longleftrightarrow \mathbf{V} = V_m \angle \phi$$

$$\sin\theta = \cos(\theta - 90^\circ), \quad \omega = 2\pi f, \quad T = 1/f$$

$$Z_R = R, \quad Z_L = j\omega L, \quad Z_C = 1/(j\omega C)$$

Keep peak magnitudes. Add impedances in series. For PS8's series RLC circuit, X = ωL − 1/(ωC).

Solve for the requested current or voltage

Series RL: $Z = R + j\omega L, \qquad \mathbf{I} = \mathbf{V}_s/Z$

$$\mathbf{V}_R = R\mathbf{I}_R, \quad \mathbf{V}_L = j\omega L \mathbf{I}_L, \quad Z = \mathbf{V}/\mathbf{I}$$

Parallel RL: $\mathbf{I}_s = \mathbf{V}/R + \mathbf{V}/(j\omega L)$

For Z = R + jX, |Z| = √(R² + X²) and θ = atan2(X,R). Divide magnitudes, subtract angles. $I_m \angle \beta$ gives i(t) = I_m cos(ωt + β).

PS 6–8 Related problem-set formulas

DC switching and stored energy

$$i_C = C\, dv_C/dt, \quad v_C(t) = v_C(t_0) + \tfrac{1}{C}\int_{t_0}^{t} i_C(u)\, du$$

$$w_L = \tfrac{1}{2} L i_L^2, \quad w_R = \int_{t_1}^{t_2} R i_R^2\, dt$$

$$\tau_{RC} = R_{\text{Th}} C, \quad v_C(t) = V_f + (V_0 - V_f)e^{-t/\tau_{RC}}$$

Mixed AC branches

$$Z_p = \frac{Z_1 Z_2}{Z_1 + Z_2}, \quad \mathbf{V}_x = \mathbf{V}_s \frac{Z_x}{\sum Z_{\text{series}}}$$

$$\mathbf{I}_1 = \mathbf{I}_s \frac{Z_2}{Z_1 + Z_2} \quad \text{(two parallel branches)}$$